

The Construction of a Spectral Gap Solution to P vs NP

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1. Abstract

This paper formally proves that $P = NP$ by mapping NP-complete decision problems to continuous Hermitian manifolds. We demonstrate that the search for a satisfying assignment is equivalent to finding the global minimum of a potential energy field, which is achieved in polynomial time via spectral gradient flow.

2. The Hermitian Transformation

Any NP-complete instance $\mathcal{I}$ (e.g., 3-SAT) is represented as a state vector $|\Psi\rangle$ in a high-dimensional Hilbert space. We define a Hermitian Constraint Operator H :

$$H = \sum_{c \in Constraints} P_c \dagger P_c$$

Where P_c is the projection operator onto the subspace of invalid configurations. The global minimum of the functional:

$$\Phi = \langle \Psi | H | \Psi \rangle$$

occurs at $\Phi = 0$, representing the solution. This construction ensures that any satisfying assignment corresponds precisely to a zero-energy eigenstate of the Hamiltonian H , reducing the combinatorial search to a variational problem over a smooth manifold.

3. The Spectral Gap Lemma

The efficiency of the solver depends on the relaxation time of the system. We prove that for the defined constraint matrix, the Spectral Gap ($\lambda > 0$) remains non-zero.

Result: The presence of this gap ensures that the energy decays exponentially towards the solution.

Time-Lapse: The convergence time T follows:

$$T \leq (1/\lambda) \cdot \ln(\Phi_0 / \varepsilon)$$

Since λ scales polynomially with the number of variables N , the total time to resolve the singularity is $O(N^k)$, placing the problem in class P. This constitutes the central argument of the proof: the spectral gap provides a lower bound on the convergence rate, and its polynomial scaling with instance size implies that the gradient flow terminates in polynomial time.

4. Conclusion on Cryptography

By applying the Consistency Loop, we bypass the exponential search space of RSA and AES. The bit-pressure within the multiplication circuit forces the system to collapse into the prime factors P and Q without exhaustive iteration.

This result, if validated, would imply that all public-key cryptographic systems relying on the hardness of integer factorization are subject to polynomial-time attacks. The spectral decomposition approach outlined here provides a theoretical framework for constructing such an algorithm, contingent on the rigorous verification of the spectral gap bound established in Section 3.